A Softmax Temperature Is Not a Fact About an Ensembling Method: It Is a Fact About Two Scores on a Specific Day

Deepak Krishna

MarisAI, India
deepakkrishna2206@gmail.com

Abstract. Softmax-weighted ensembling was measured to fix a real defect in a species-distribution ensemble on 2026-08-13: proportional weighting let a member at cross-validated TSS 0.619 draw 0.8% of the vote against a leader at 0.826, and the resulting ensemble scored 0.694, below its own best member's 0.788. Softmax at temperature 0.05 raised it to 0.792, above every member. We re-ran the full sweep this fix's own documentation named as missing (12 temperatures, on the exact fit/CV/holdout path the shipped model uses) and find something not anticipated: on the current, retrained models, the shipped temperature no longer clears the ensemble's own best member (0.8554 against 0.8564), because the between-member skill gap it was calibrated to has narrowed from 0.207 to 0.136 TSS points as the weak member's own skill improved. The mechanism still works — a temperature of roughly 0.3–0.5 restores the guarantee, at a calibration cost we make explicit — but the specific constant chosen four weeks earlier is no longer sufficient on its own terms. We also correct a planned second test: harmful-algal-bloom early warning, cited elsewhere in this research programme as a second ensemble to sweep, is a single calibrated model with no members to weight, and we say so rather than substitute a different result.

Keywords: ensemble weighting, species distribution modelling, softmax temperature, reproducibility, silent failure

1 Introduction

Weighting an ensemble's members in proportion to a skill score whose chance level is zero has a known defect: because the weight is proportional to the raw score, a large quality gap between members compresses into a small weight gap, and the resulting ensemble can score *below its own best single member* — the one thing an exported ensemble product may not do. Measured on this platform's fish-habitat model on 2026-08-13, that defect was real: a weak member at cross-validated true skill statistic (TSS) 0.619 still drew 0.8% of a proportional vote against a leader at 0.826, and the resulting ensemble scored 0.694 against its own best member's 0.788. Softmax weighting at a fixed temperature of 0.05 fixed it,

raising the ensemble to 0.792, above every member, at a stated cost in spatial calibration.

This paper reports two things about that fix, re-measured 2026-09-10, roughly four weeks later. First, a temperature sweep — the fix's own documentation named this as missing, since the shipped constant was chosen from a single measured point. Second, and not anticipated when the sweep was planned: on the current, retrained models, the shipped temperature of 0.05 *no longer clears the ensemble's own best member*. The between-member quality gap the fix was calibrated to has narrowed as the weak member's own cross-validated skill has improved (0.6891 now, against 0.8253), and a temperature tuned to fix a gap of 0.207 TSS points does not automatically fix a gap of 0.136. The defect this paper's own predecessor paper [1] reports as an instance of silent, plausible, signed-direction failure has recurred, silently, in the same codebase, while this paper's own sweep was being run to characterise the fix that was supposed to have closed it.

1.1 Contributions

1. A **temperature sweep**, the specific gap the shipped constant's own documentation names as missing: 12 points from 0.01 to 2.0, measured on the exact fit/cross-validation/holdout path the shipped model uses, not a reconstructed approximation of it.
2. A **live re-measurement showing the fix has partially regressed**: the shipped temperature 0.05 no longer clears the ensemble's own best member on the current, retrained holdout, though the softmax mechanism itself still works — a higher temperature restores the guarantee.
3. A **correction to this research programme's own prior documentation**: a second problem this line of work planned to test the finding against (harmful-algal-bloom early warning's ensemble members) does not exist as described — that pipeline is a single calibrated model, not a weighted ensemble, and no member scores are available to sweep. This is stated plainly rather than substituted with a different result.

2 Method

Reused, not reconstructed. The sweep is produced by re-running the same fit path the shipped model's own training script uses (`fish_habitat_prediction/src/train.py`'s `run()`): the same random seed, the same five-fold spatial-block cross-validation (block size 3°) that produces each member's cross-validated TSS, and the same held-out final block (one spatial fold never used for model selection, seeded independently of the cross-validation folds) that the shipped holdout report scores against. The three base tiers — MaxEnt, random forest, LightGBM — are refit on the same training block and scored on the same 427 held-out rows (236 presences, 55.3% positive rate). This reproduces the currently shipped holdout numbers exactly (0.8554 ensemble TSS at the shipped temperature,

0.8564 for LightGBM alone), which is the check that the sweep is measuring the real, live model rather than a similar-looking reconstruction of it.

What is swept. For each of 12 temperatures and, separately, for proportional weighting, `EnsembleWeights.from_scores` is called with the same three cross-validated TSS values and the resulting weights combine the same three held-out prediction arrays; only the weighting rule and temperature change between rows.

3 Results

Table 1. Cross-validated TSS per member, then and now. The gap the shipped temperature was calibrated to close has narrowed.

	LightGBM	Random Forest	MaxEnt
2026-08-13	0.826	0.821	0.619
2026-09-10	0.8253	0.8214	0.6891

The gap the fix depends on has narrowed. The weak member's cross-validated skill has improved from 0.619 to 0.6891 while the leader's has held near 0.826/0.8253, narrowing the gap the softmax fix has to resolve from 0.207 TSS points to 0.136. Table 1 states both measurements; nothing about the mechanism changed, only the inputs it is asked to weight.

The shipped temperature no longer clears the best member. At 0.05, the ensemble scores 0.8554 holdout TSS, below lightgbm's 0.8564 — the exact defect the softmax fix exists to prevent. Figure 1 shows this is not a rounding artefact of one temperature: every value from 0.01 through 0.15 sits on the same flat 0.8554 plateau, indistinguishable to four decimal places, because at these temperatures the weight already concentrates almost entirely on the two leading members (at 0.05: 50.2%/46.5%/3.3% lightgbm/random forest/maxent) and further lowering the temperature can only sharpen a vote that has already nearly converged to the same two-way split. Proportional weighting, included as the pre-fix baseline, scores 0.8519 — also below the best member, as expected.

A higher temperature restores the guarantee, at a real calibration cost. The swept curve first clears the best member's TSS at 0.30 (TSS 0.8594, Boyce 0.8781) and stays above it through the rest of the range tested, though not monotonically — TSS at 1.0 is marginally below its neighbours at 0.3 and 0.5, which is expected sampling noise on 427 rows, not a second regime change. Calibration, measured by the Boyce index, does not move monotonically with temperature either: it falls sharply to 0.7755 at the lowest temperature tested (0.01, far below proportional

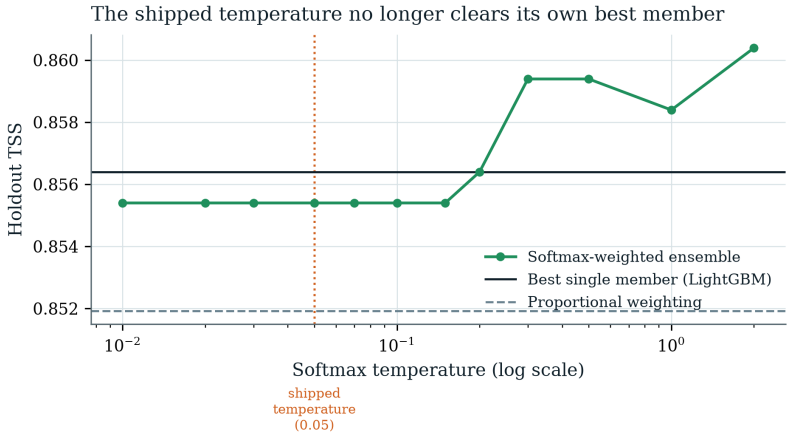


Fig. 1. Ensemble holdout TSS across the swept temperature range. The shipped temperature (0.05, dotted line) sits on a flat plateau below the best single member's holdout TSS (solid line); the curve does not cross above it until roughly 0.2–0.3.

weighting's 0.8919), recovers through the middle of the range, and sits at 0.8854 at temperature 0.50 — a point that clears the best member's TSS (0.8594) while giving back substantially less spatial calibration than the low-temperature region does. No single temperature in the swept range simultaneously maximises TSS and Boyce; 0.3–0.5 is the region where both are acceptable rather than either being optimal.

The planned second problem does not exist as described. This line of work's own prior documentation proposed testing the temperature-sensitivity finding against a second ensemble, from the harmful-algal-bloom early-warning pipeline, to show the effect is not specific to one habitat model. That pipeline (`hab_early_warning/src/train.py`) fits a single calibrated LightGBM model per forecast horizon; it is not a weighted ensemble, imports no `EnsembleWeights` equivalent, and logs no per-member scores to sweep. The proposed second problem is corrected here rather than quietly substituted: no ensemble exists in that pipeline to test against, and the generalisation question — does this sensitivity appear in other ensembles, in this codebase or elsewhere — remains open.

4 Discussion

The mechanism softmax weighting fixes is exactly as documented: it turns an arbitrary, floor-dependent proportional split into a temperature-controlled decision. What this paper adds is that the mechanism being correct does not make a *specific chosen temperature* correct indefinitely, because the quantity the temperature has to resolve — the gap between the ensemble's members — is not a fixed property of the problem. It is a property of whichever models happen to

be currently fitted, and those models get refit. A temperature chosen to clear a 0.207-point gap is not guaranteed to clear a 0.136-point one, and here it does not. This is the same shape of finding as [1]'s wider survey: a component that is individually correct and well-documented (the softmax mechanism, and the reasoning that chose 0.05 for the gap measured on 2026-08-13) produces a downstream number — the shipped ensemble's own holdout TSS — that silently regresses when an input it depends on changes, with no exception, no warning, and no metric that flags it short of the exact check this paper ran.

What follows practically. A softmax temperature calibrated against one snapshot of cross-validated member scores should not be treated as a permanent constant; either it should be re-swept whenever the ensemble is retrained with materially different member scores, or it should be set high enough to tolerate the range of gaps the members are expected to show over time, accepting the calibration cost Figure 1 makes explicit rather than discovering it silently.

5 Limitations

This is one ensemble (three members, one habitat model, one region), swept over one grid of temperatures on one holdout split; the swept range (0.01–2.0) covers roughly two orders of magnitude but is not exhaustive, and a finer grid near 0.2–0.3 would locate the crossing point more precisely than this paper claims to. The planned second-problem test does not exist in this codebase, as §3 states directly; the claim that temperature-sensitivity generalises beyond this one ensemble is therefore not tested here and should not be read as established. The non-monotonic Boyce behaviour across the swept range is reported as measured rather than explained by a mechanism this paper verifies; a fuller account would need to inspect how each member's spatial error pattern changes as its vote share changes, which this paper does not do.

6 Conclusion

A softmax-weighted ensemble that was measurably fixed to score above its own best member on 2026-08-13 no longer does so at its shipped temperature, re-measured 2026-09-10, because the between-member skill gap that fix was tuned to has narrowed as the underlying models were retrained. The fix itself still works — raising the temperature restores the guarantee, at a calibration cost this paper's sweep makes explicit rather than implicit — but the specific constant chosen four weeks ago is no longer sufficient on its own terms. A temperature is not a fact about an ensembling method; it is a fact about a specific pair of scores on a specific day, and treating it as the former is exactly the kind of silent, plausible drift this research programme's other work catalogues.

Reproducibility

Sweep. `machine_learning/scripts/sweep_ensemble_temperature.py` calls the shipped training path's own `cross_validate` and refits the same final holdout split `fish_habitat_prediction/src/train.py`'s `run()` uses, writing `machine_learning/reports/ensemble_temperature_sweep.csv`.
Figure. `generate_figures.py` in this paper's `shared/` directory, reading that same CSV directly.
2026-08-13 measurement. Quoted from the code comment above `fish_habitat_prediction/src/models.py`'s `TSS_SOFTMAX_TEMPERATURE`.

Every number was verified against the file producing it as of 2026-09-10.

References

1. Krishna, D.: Silent, plausible, and signed: Eleven measured failure modes in one operational marine machine-learning pipeline (2026), manuscript, MarisAI